

UNIT 1

NUMBERS AND ARITHMETIC OPERATIONS



PLACE VALUE

Just as each digit has a place in a number, the digit also has value based on its place. This is called place value.

Thousands			Ones		
H-Th	T-Th	Th	H	T	O
0	0	0	0	0	0

Thousands	Ones
768	172

Here,

H-TH stands for hundred thousand

T-TH stands for ten thousand

TH stands for thousand

H stands for hundred

T stands for tens

O stands for ones

Look at number 768,172

Example 1:

Write 567,812 in words.

Thousands			Ones		
H-Th	T-Th	Th	H	T	O
5	6	7	8	1	2

Answer: Five hundred sixty seven thousands and eight hundred.

EXERCISE 1.1

Q1. Separate by periods and read the following numbers.

1) 45672 = _____

2) 54801 = _____

3) 2670273 = _____

4) 4536580 = _____

5) 34296127 = _____

6) 100000000 = _____

7) 9923456320 = _____

8) 6123450238 = _____

Q2. Fill in the correct thousands, hundreds, tens and ones for the given number.

Number	Thousands, Hundreds, Tens and Ones
6723	_____ thousands _____ hundreds _____ tens _____ ones
8954	_____ thousands _____ hundreds _____ tens _____ ones
2375	_____ thousands _____ hundreds _____ tens _____ ones

5698	_____ thousands
	_____ hundreds
	_____ tens
	_____ ones

Q3. Solve the following.

1) Write the place value of the underlined digit under each of the numbers.

27,<u>5</u>02	<u>7</u>1,918	132,<u>8</u>25	7<u>4</u>9,327	28,<u>1</u>76
500				

<u>5</u>13,295	8<u>3</u>4,247	36,<u>4</u>2<u>9</u>	<u>6</u>25,231	<u>9</u>17,438

2) Write these numbers in expanded form.

13,459	$10,000 + 3,000 + 400 + 50 + 9$
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35,916	
---------------	--

132,756	
----------------	--

849,018	
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3) Write these numbers in standard form.

$$10,000 + 3,000 + 500 + 80 + 2$$

13,582

$$80,000 + 7,000 + 600 + 90 + 5$$

$$100,000 + 40,000 + 9,000 + 400 + 50 + 3$$

$$200,000 + 60,000 + 800 + 70 + 4$$

$$600,000 + 9,000 + 400 + 90$$

4) Fill in the missing parts in these numbers

1) 27,582 = _____ thousands _____ hundreds _____
tens _____ ones

2) 38,214 = _____ thousands _____ hundreds _____
tens _____ ones

3) 135,634 = _____ thousands _____ hundreds _____
tens _____ ones

Q4. Write the numbers according to their place value.

	Ten Thousands	Thousands	Hundreds	Tens	Units
1) 38,625	30000	8000	600	20	5
2) 40,863					
3) 40,581					

4) 64,536

5) 73,966

6) 75,241

Q5. Complete each equation to make it true.

1) 2 thousands + 5 hundreds + 2 tens + 2 ones = _____

2) 3 thousands + 3 hundreds + 7 tens + 9 ones = _____

3) 4 thousands + 3 hundreds + 6 tens + 1 one = _____

4) 5 thousands + 2 hundreds + 0 tens + 7 ones = _____

5) 1 thousand + 3 hundred + 7 tens + 4 ones = _____

ADDITION**Example 1:**

Add 638,941 and 347,036

H-Th	T-Th	Th	H	T	O
6	^① 3	8	9	4	1
+ 3	4	7	0	3	6
9	8	5	9	7	7

Example 2:

Add 359,990 and 406,780

H-Th	T-Th	Th	H	T	O
3	^① 5	^① 9	^① 9	9	0
+ 4	0	6	7	8	0
7	6	6	7	7	0

Example 3:

Add 5,389,624 and 930,157.

$$\begin{array}{r}
 \textcircled{1} 5 \quad \textcircled{1} 3 \quad 8 \quad 9 \quad 6 \quad \textcircled{1} 2 \quad 4 \\
 + \quad 9 \quad 3 \quad 0 \quad 1 \quad 5 \quad 7 \\
 \hline
 6 \quad 3 \quad 1 \quad 9 \quad 7 \quad 8 \quad 1
 \end{array}$$

EXERCISE 1.2

Q1. Add the following.

1)
$$\begin{array}{r}
 6 \ 0 \ 4 \ 5 \ 1 \ 7 \ 3 \\
 + \ 2 \ 1 \ 4 \ 6 \ 1 \ 8 \ 5 \\
 \hline
 \end{array}$$

2)
$$\begin{array}{r}
 1 \ 2 \ 7 \ 4 \ 5 \ 3 \ 8 \\
 + \ 6 \ 3 \ 8 \ 2 \ 7 \ 1 \ 4 \\
 \hline
 \end{array}$$

3)
$$\begin{array}{r}
 7 \ 5 \ 3 \ 2 \ 9 \ 6 \ 1 \\
 + \ 6 \ 7 \ 3 \ 8 \ 2 \ 9 \\
 \hline
 \end{array}$$

4)
$$\begin{array}{r}
 9 \ 0 \ 2 \ 4 \ 3 \ 8 \ 2 \\
 + \ 5 \ 8 \ 3 \ 9 \ 2 \ 0 \ 7 \\
 \hline
 \end{array}$$

5)
$$\begin{array}{r}
 6 \ 4 \ 7 \ 3 \ 2 \ 9 \ 1 \\
 + \ 8 \ 3 \ 7 \ 2 \ 9 \ 1 \ 8 \\
 \hline
 \end{array}$$

6)
$$\begin{array}{r}
 7 \ 3 \ 8 \ 4 \ 7 \ 3 \ 5 \\
 + \ 8 \ 3 \ 7 \ 2 \ 6 \ 1 \ 3 \\
 \hline
 \end{array}$$

7)
$$\begin{array}{r}
 4 \ 0 \ 3 \ 8 \ 2 \ 9 \ 1 \\
 + \ 6 \ 3 \ 7 \ 2 \ 8 \ 2 \\
 \hline
 \end{array}$$

8)
$$\begin{array}{r}
 8 \ 9 \ 3 \ 8 \ 2 \ 7 \ 1 \\
 + \ 5 \ 0 \ 9 \ 3 \ 8 \ 7 \ 2 \\
 \hline
 \end{array}$$

Q2. Add the following.

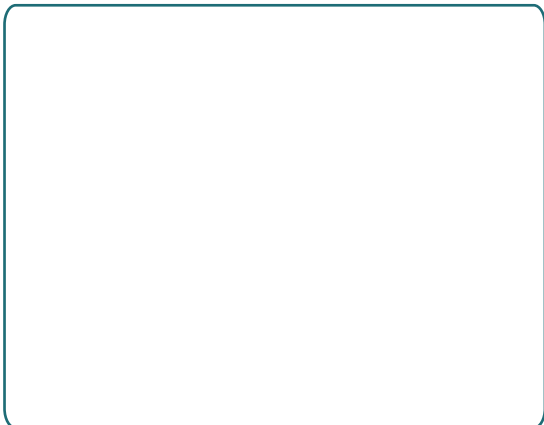
1) 680,563 and 563,168



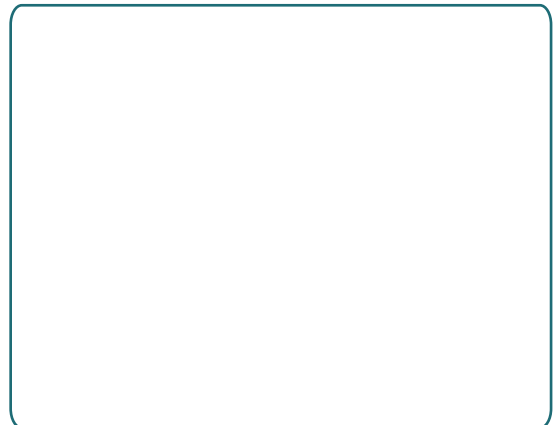
2) 3,541,371 and 232,164



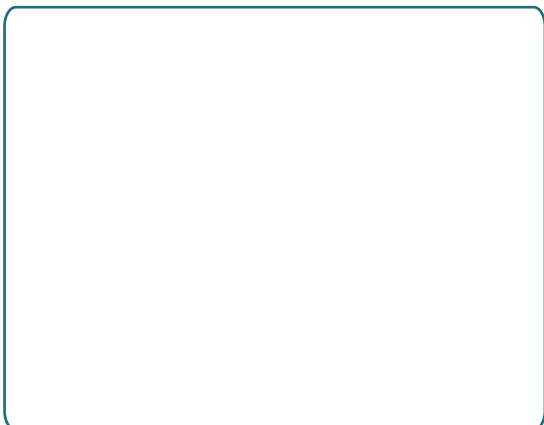
3) 34,399,188 and 6,412,508



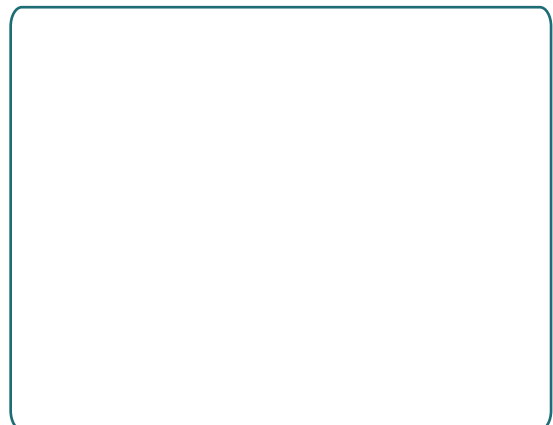
4) 65,943,022 and 22,913,9



5) 840,233,419 and 65,113,846



6) 733,050,195 and 188,439,919



Q3. Fill the missing digits in the following.

1)
$$\begin{array}{r} 5 \ 3 \ 6 \ 2 \ \bigcirc \ 4 \\ + 3 \ \bigcirc \ 3 \ \bigcirc \ 2 \ 2 \\ \hline 8 \ 8 \ 9 \ 6 \ 3 \ 6 \end{array}$$

2)
$$\begin{array}{r} 3 \ 4 \ 7 \ 8 \ 4 \ 5 \\ + 4 \ \bigcirc \ 2 \ \bigcirc \ 1 \ \bigcirc \\ \hline 7 \ 7 \ 0 \ 1 \ 5 \ 9 \end{array}$$

3)
$$\begin{array}{r} 5 \ 6 \ 0 \ 1 \ 8 \ 9 \ 7 \\ + 2 \ \bigcirc \ 5 \ \bigcirc \ 4 \ \bigcirc \ 1 \\ \hline 8 \ 1 \ 5 \ 5 \ 3 \ 1 \ 8 \end{array}$$

4)
$$\begin{array}{r} 3 \ 4 \ \bigcirc \ 5 \ \bigcirc \ 3 \ 7 \\ + 2 \ \bigcirc \ 4 \ 2 \ 0 \ \bigcirc \ 2 \\ \hline 6 \ 0 \ 6 \ 7 \ 1 \ 4 \ 9 \end{array}$$

SUBTRACTION

Example 1:

Subtract 430,912 from 871,032.

$$\begin{array}{r} 8 \quad 7 \quad \textcircled{0}1 \quad \textcircled{10}0 \quad 3 \quad 2 \\ - 4 \quad 3 \quad 0 \quad 9 \quad 1 \quad 2 \\ \hline 4 \quad 4 \quad 0 \quad 1 \quad 2 \quad 0 \end{array}$$

Example 2:

Subtract 273,587 from 307,843.

$$\begin{array}{r} 3 \quad \textcircled{10}0 \quad 7 \quad \textcircled{7}8 \quad \textcircled{13}4 \quad \textcircled{13}3 \\ - 2 \quad 7 \quad 3 \quad 5 \quad 8 \quad 7 \\ \hline 0 \quad 3 \quad 4 \quad 2 \quad 5 \quad 6 \end{array}$$

EXERCISE 1.3

Q1. Subtract the following.

$$\begin{array}{r} 1) \quad 5 \ 0 \ 7 \ 5 \\ - \quad 2 \ 3 \ 1 \ 4 \\ \hline \end{array}$$

$$\begin{array}{r} 2) \quad 6 \ 7 \ 5 \ 2 \\ - \quad 3 \ 8 \ 2 \ 7 \\ \hline \end{array}$$

$$\begin{array}{r} 3) \quad 4 \ 2 \ 9 \ 5 \\ - \quad 3 \ 0 \ 7 \ 9 \\ \hline \end{array}$$

$$\begin{array}{r} 4) \quad 3 \ 4 \ 2 \ 6 \\ - \quad 2 \ 2 \ 3 \ 9 \\ \hline \end{array}$$

$$\begin{array}{r} 5) \quad 8 \ 7 \ 2 \ 1 \\ - \quad 5 \ 3 \ 7 \ 6 \\ \hline \end{array}$$

$$\begin{array}{r} 6) \quad 9 \ 0 \ 2 \ 7 \\ - \quad 4 \ 6 \ 5 \\ \hline \end{array}$$

$$\begin{array}{r} 7) \quad 6 \ 5 \ 3 \ 8 \\ - \quad 1 \ 8 \ 6 \ 5 \\ \hline \end{array}$$

$$\begin{array}{r} 8) \quad 7 \ 5 \ 6 \ 0 \\ - \quad 2 \ 6 \ 3 \ 4 \\ \hline \end{array}$$

$$\begin{array}{r} 9) \quad 5 \ 3 \ 6 \ 8 \\ - \quad 9 \ 4 \ 7 \\ \hline \end{array}$$

$$\begin{array}{r} 10) \quad 7 \ 1 \ 3 \ 5 \\ - \quad 2 \ 8 \ 6 \ 4 \\ \hline \end{array}$$

$$\begin{array}{r} 11) \quad 8 \ 6 \ 5 \ 6 \\ - \quad 6 \ 4 \ 5 \ 1 \\ \hline \end{array}$$

$$\begin{array}{r} 12) \quad 6 \ 0 \ 5 \ 7 \\ - \quad 3 \ 2 \ 4 \ 9 \\ \hline \end{array}$$

Q2. Solve:

$$\begin{array}{r} 1) \quad 4 \ 7 \ 8 \ 3 \ 5 \\ - \quad 2 \ 4 \ 5 \ 1 \\ \hline \end{array}$$

$$\begin{array}{r} 2) \quad 7 \ 3 \ 8 \ 2 \ 6 \\ - \quad 1 \ 5 \ 0 \ 8 \\ \hline \end{array}$$

$$\begin{array}{r} 3) \quad 9 \ 5 \ 2 \ 7 \ 6 \\ - \quad 3 \ 4 \ 6 \ 1 \\ \hline \end{array}$$

$$\begin{array}{r} 4) \quad 7 \ 1 \ 6 \ 8 \ 5 \\ - \quad 4 \ 2 \ 3 \ 3 \\ \hline \end{array}$$

$$\begin{array}{r} 5) \quad 8 \ 3 \ 6 \ 9 \ 1 \\ - \quad 4 \ 1 \ 0 \ 4 \ 5 \\ \hline \end{array}$$

$$\begin{array}{r} 6) \quad 6 \ 9 \ 3 \ 6 \ 6 \\ - \quad 1 \ 6 \ 7 \ 2 \ 5 \\ \hline \end{array}$$

Q3. Subtract the following.

1) 214, 379 from 600,500

2) 856,394 from 3,767,595

3) 4,930,109 from 5,851,036

4) 5,396,138 from 43,547,967

6) 134,258,369 from 656,148,154

[illegible]

Q1. Multiply the given numbers.

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2) 3 4 , 5 6 9 x 1 0

$$\begin{array}{r} 4136 \\ \times 10 \\ \hline \end{array}$$

	3	4	5	6	9
		x		10	
+					

3) 15,347 x 100

	1	5	3	4	7
	x	1	0	0	
+					

2) 27,796 x 100

	2	7	7	9	6
	x	1	0	0	
+					

Q3. Multiply.

1) 25839 x 33

2) 243419 x 86

3) 65204 x 75

4) 467808 x 92

Step 1: Since digit of highest place value 6 is less than divisor 24. So, we take 64 multiply 24 by 2. Write 2 as quotient and subtract ($64 - 48 = 16$)

Step 2: 1 is the remainder so we get 163. Multiply 24 by 6 and get 144. Write 6 as quotient and subtract ($163 - 144 = 19$).

Step 3: The next digit in dividend is 1. So, we place it after 19, we get 191. Multiply 24 by 7. Write 7 as quotient and subtract ($191 - 168 = 23$).

Step 4: Next digit in dividend is 8, we place it after remainder 23. We get 238. Multiply 24 by 9, we get 216. Write 9 as quotient and subtract ($238 - 216 = 22$).

Step 5: Next digit in dividend is 5, so we put it after remainder 22 and get 225. Multiply 24 by 9, we get 216. Write 9 as quotient and subtract ($225 - 216 = 9$).

Example 2.

Divide 837,576 by 123.

$$\begin{array}{r}
 6809 \\
 123 \overline{) 837576} \\
 \underline{- 738} \\
 0995 \\
 \underline{- 984} \\
 01176 \\
 \underline{- 1107} \\
 69
 \end{array}$$

Hence quotient is 6809 and remainder is 69.

EXERCISE 1.4

Q1. Solve the following.

- | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|
| 1) $423 \div 8$ | 2) $674 \div 6$ | 3) $954 \div 5$ | 4) $816 \div 7$ |
| 5) $562 \div 4$ | 6) $785 \div 9$ | 7) $8218 \div 14$ | 8) $748 \div 15$ |
| 9) $635 \div 16$ | 10) $582 \div 17$ | 11) $1763 \div 18$ | 12) $4826 \div 19$ |
| 13) $5893 \div 23$ | 14) $3168 \div 24$ | 15) $7224 \div 24$ | 16) $6214 \div 26$ |
| 17) $1715 \div 35$ | 18) $5550 \div 37$ | 19) $2665 \div 41$ | 20) $1313 \div 43$ |

WORD PROBLEMS

- 1 Fouzia had 145,320 rupees. Her father gave her 54,304 rupees more. How many rupees does Fouzia have altogether?
- 2 Sobia gave a 5,000 rupee note to the shopkeeper for purchasing an electric doll worth Rs 2,300. Find the remaining amount that Sobia received.
- 3 A school collected funds for flood relief. 225 students of the school contributed equally, amounting to Rs 25,650 in total. Find the contribution of each student.
- 4 A factory produces 235,806 floor marble tiles in a day. How many tiles will be produced in 32 days?
- 5 A container holds 9,475 litres of milk. How many litres of milk will 354 such containers hold?
- 6 Find the number of rolls required to pack 531,675 metres of cloth. Each roll contains 45 metres.

BODMAS RULE

The rule of **BODMAS** is used to remembering the order of preferences of mathematical operations.

B	Brackets	Solve within brackets ()
O	of	to multiply
D	Divide	$\div$
M	Multiply	$\times$
A	Add	$+$
S	Subtract	$-$

Preference of solving brackets is as follows:

- (i) () parentheses
- (ii) { } curly brackets
- (iii) [] square brackets.

Brackets () are known as parentheses.

Example 1:

Solve by using BODMAS rule $135 \div 15 + 6 - 5 \times 2$

Solution: $135 \div 15 + 6 - 5 \times 2$

- (By applying DMAS rule we first perform **division**).

$$= 9 + 6 - 5 \times 2$$

$$(135 \div 15 = 9)$$

- Then perform **multiplication**. ($5 \times 2 = 10$).

$$= 9 + 6 - 10$$

- In the last perform **subtraction**.

$$= 15 - 10$$

$$= 5$$

Example 2:**Solve by using BODMAS rule $64 - (6 \times 2) \times 3$** **Solution:**

$$= 64 - (6 \times 2) \times 3$$
 First perform operation of 6×2

$$= 64 - 12 \times 3$$
 Then remove bracket after **Multiplication**

$$= 64 - 36$$
 and then **Subtract**

$$= 28$$

Example 3:**Solve $82 - 32 \times 2 + (8 + 20 \div 40)$** **Solution:**

$$82 - 32 \times 2 + (8 + 20 \div 40)$$

$$= 82 - 64 + (8 + 20 \div 40)$$
 (Now perform division)

$$= 82 - 64 + (8 + 5)$$
 (Now remove bracket)

$$= 82 - 64 + 13$$
 (In the last perform addition and subtraction)

$$= 18 + 13$$

$$= 31$$

EXERCISE 1.5

Q1. Solve by BODMAS rule.

1) $(12 - 3) \times (12 - 6)$ **2)** $3 + 5 + (10 \div 2)$ **3)** $(40 \div 5) \times 5 \div 5$

4) $(9 - 3) + 10 \times 4$ **5)** $8 + 20 - (4 \times 3)$ **6)** $5 \times 6 + (9 - 3)$

Q2. Find the answer to each equation.

1) $(10 \times 7 + 40) + 38 \times 10 =$ _____

2) $31 - 1 \times 5 - 8 + 33 =$ _____

3) $(10 \times 7 + 40) + 38 \times 10 =$ _____

4) $31 - 1 \times 5 - 8 + 33 =$ _____

5) $31 - 1 \times 5 - 8 + 33 =$ _____

Q3. Simplify the following.

1) $30 \div 6 + 10 - 2 \times 5$

2) $48 \div 16 \times 2 + 17 - 9$

3) $(18 + 10) - (3 \times 6)$

4) $(10 \times 8) \div (20 \div 5)$

5) $\{20 - (24 - 10)\} + 7$

6) $\{20 + (15 + 5)\} \div 5$

7) $12 \times 3 \div 3 \times 4 - 2 + 6$